

3 Network Metrics

For large networks we can't usually just plot them or examine the individual nodes. Just like for non-relational data we use various summary statistics to help us describe large networks or identify interesting nodes.

Reading

[1] Chapter 1, 7.3, [2] Chapters 7,8.

3.1 Clustering coefficient

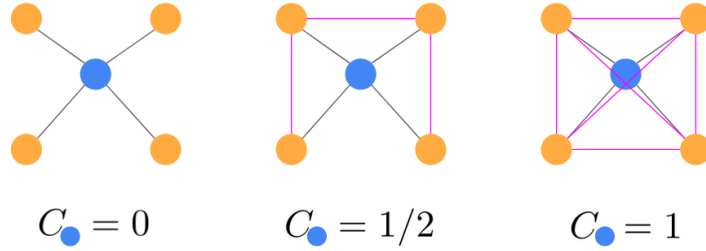


Figure 1: Clustering coefficients of a node of degree 4 in different neighbourhoods.

The **clustering coefficient** helps us to identify how dense connections are in the neighbourhood of a node. A node i has k_i neighbours. Those k_i neighbours have L_i connections between themselves. The maximum number they could have is $\binom{k_i}{2} = \frac{k_i(k_i-1)}{2}$. The clustering coefficient of i is the ratio of actual over potential connections between neighbours of i ,

$$C_i = \frac{2L_i}{k_i(k_i - 1)} \quad (1)$$

Using the fact that powers of the adjacency matrix count paths, we can write an expression for L_i in terms of the adjacency matrix. If $k_i \leq 1$ we just define C_i to be 0. Otherwise neighbours j and k of i are neighbours of each other if $A_{jk} = 1$. Thus there is a path of length 3 from i to j to k to i . These paths are counted by the cube of the adjacency matrix

$$2L_i = \sum_{j,k} A_{ij} A_{jk} A_{ki} \quad (2)$$

The two comes because every path is counted twice ijk and kji , for example. It is sometimes useful to compute the **average clustering coefficient**

$$C = \frac{1}{N} \sum_i C_i \quad (3)$$

3.2 Global clustering coefficient

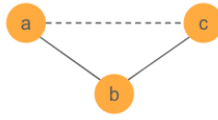


Figure 2: Triplet abc is closed if the dashed line is an edge, if not, it is open.

There is another definition of clustering coefficient which is quite similar to the above. A **triplet** is three nodes a, b, c which are connected, a to b , b to c . If the vertices form a triangle i.e. c connected to a , then the triplet is **closed**, otherwise it is **open**. The **global clustering coefficient** is

$$C_{\Delta} = \frac{\text{number of closed triplets}}{\text{number of all triplets}} \quad (4)$$

Because a triangle is formed of three closed triplets (abc, bca, cab) the numerator can be rewritten

$$C_{\Delta} = \frac{3 \times \text{number of triangles}}{\text{number of all triplets}} \quad (5)$$

You can think of this as the fraction triangles in the actual network over the fraction that could be in the network.

Using similar logic as for the local clustering coefficient, the number of closed triplets is

$$\sum_{ijk} A_{ij} A_{jk} A_{ki}$$

If i has k_i neighbours we can choose one of them to be the left partner leaving $k_i - 1$ for the right partner, so the number of all triplets is

$$\frac{1}{2} \sum_i k_i (k_i - 1)$$

we then divide by two because we've counted every triplet twice. Together this means

$$C_{\Delta} = \frac{\sum_{ijk} A_{ij} A_{jk} A_{ki}}{\frac{1}{2} \sum_i k_i (k_i - 1)} \quad (6)$$

versus

$$C = \frac{1}{N} \sum_i \frac{\sum_{jk} A_{ij} A_{jk} A_{ki}}{k_i (k_i - 1)}$$

They are quite similar, but average over the nodes differently. The clustering coefficient is nice because it is a sum of local measurements but its average is dominated by low degree nodes, so C_{Δ} might be a better measure of global clustering.

3.3 Degree centrality

An important question in the analysis of a network is often *which is the most important node in the network?*. Different measures of **centrality** attempt to quantify this. The simplest is **degree centrality** which is just k_i i.e. the most important nodes are the ones that are the most connected. Though simple, this can often be quite a useful way of sorting the nodes!

3.4 Eigenvector centrality

An alternative way to determine importance is to say *important nodes are connected to important nodes*. Mathematically if the 'importance' of a node is x_i then the above is equivalent to

$$x_i = \sum_j A_{ij} x_j \quad (7)$$

where the right hand side sums the importances of all the nodes that i is connected to. If we say that the importance of a node is *proportional* to the sum of the importances of its neighbours then

$$\lambda x_i = \sum_j A_{ij} x_j \quad (8)$$

which states that $\vec{x} = (x_1, x_2, \dots, x_N)$ is an eigenvector of the adjacency matrix. Of course there are many eigenvalues and eigenvectors so we need to pick one. If we require that all the importances are non-negative, then the Perron-Frobenius theorem from linear algebra tells us that the eigenvector corresponding to the largest eigenvalue (and only this one) has this property.

Since eigenvectors are only defined up to a constant we can normalise them, for example by requiring

$$\sum_i x_i = N$$

We are guaranteed that the entries are all non-negative, but some may be zero. If this is undesirable we can modify the eigenvalue equation

$$\lambda x_i = \sum_j A_{ij} x_j + \beta_i$$

Where β_i is a small factor that ensures every importance is non-zero. This is called **Katz centrality**.

3.5 PageRank

Another common centrality metric is the PageRank score. This was used in the early version of the Google search algorithm and is usually computed for directed networks (like hyperlinks on the web). PageRank corrects an undesirable feature of eigenvector centrality - highly eigenvector-central nodes ‘transmit’ their high centrality to their neighbours. e.g. if i is a very important node with large x_i and j is another node that is connected to i and only i then

$$x_j = \sum_k A_{jk} x_k = x_i$$

so j is just as important as i . The intuition behind the PageRank comes from the idea of a ‘random surfer’ clicking on links. The random surfer has some probability α of following a link and some probability β_i of jumping to a random page (hence β is sometimes referred to as the *teleportation* factor). As the surfer clicks on links they generate a probability distribution to land on any particular page and the values of that distribution give the PageRank.

However, it is equivalent (and simpler) to view PageRank as a modified form of eigenvector centrality, where we add damping factor of $\frac{1}{k_j^{out}}$ to the importances. In this case pages which are just huge link hubs, but which are themselves not linked to by anyone, have their importance decreased. The PageRank x_i is the solution to

$$x_i = \alpha \sum_j A_{ij} \frac{x_j}{k_j^{out}} + \beta_i \quad (9)$$

Usually $\alpha = 0.85$. The vector $\vec{\beta}$ is often called the *personalization* vector. If there is no a-priori information $\beta_i = 1/N$ or $\beta_i = 1 - \alpha$ are common choices.

3.6 Closeness centrality

A different way of thinking about node importance is through path lengths. If a node is connected by short paths to all the other nodes then it is central in some sense. If d_{ij} is the shortest path between i and j then

$$l_i = \frac{1}{N} \sum_j^N d_{ij}$$

is the average distance from i to any other node. This is small for nodes which have short average distances so its inverse is large and the quantity

$$C_i = \frac{1}{l_i} \quad (10)$$

is used to measure importance. This measure is known as **closeness centrality**.

3.7 Betweenness centrality

A similar path based notion is **betweenness centrality** where a node is important if it is on a large proportion of shortest paths through the network. Such a node mediates a lot of flows on the network. Think of a city on either side of a river with one bridge, all traffic going from one side to the other has to pass over the bridge. Formally

$$b_i = \sum_{jk} \frac{n_{jk}(i)}{n_{jk}} \quad (11)$$

where n_{jk} is the number of shortest paths between j and k and $n_{jk}(i)$ is the number of those paths which pass through i . As you can guess, this is quite slow to compute for large networks.

3.8 Comparison

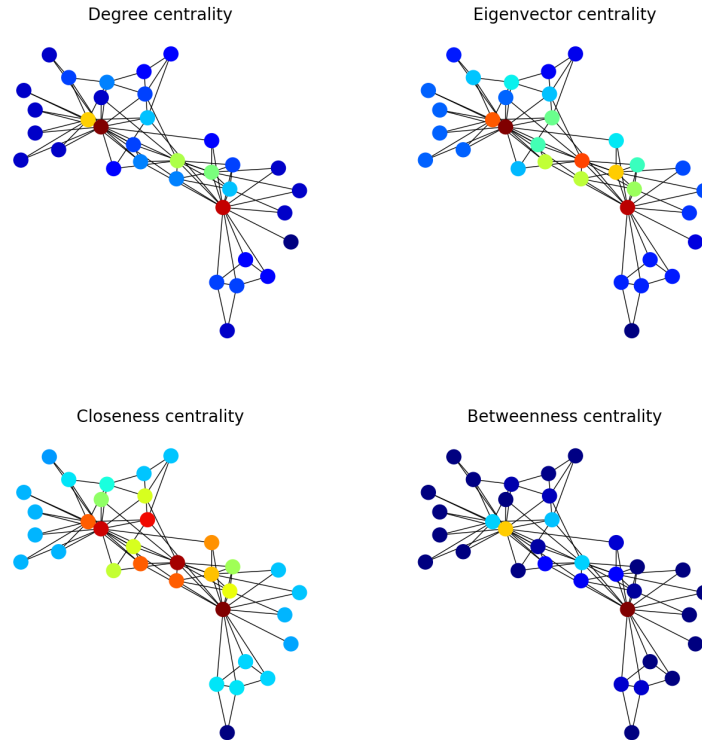


Figure 3: Different centrality measures, red is high, blue is low.

Figure 3 shows a few different centrality measures computed on the Karate Club network. They mostly identify the same nodes as important, the instructor, the president, as well as the individuals who have relationships with members of both factions.

3.9 Assortativity Coefficient

Assortativity means that similar nodes tend to be connected to each other, *like with like* and disassortativity is the opposite tendency, dissimilar nodes tend to be connected. One important property of a node is its degree, so we want a number that measures the extent to which high degree nodes connect to other high degree nodes and low degree to other low degree nodes. One way to do this is with the **assortativity coefficient**

$$r = \frac{\sum_{ij}^{k_{max}} ij(e_{ij} - q_i q_j)}{\sigma^2} \quad (12)$$

where

$$\sigma^2 = \sum_i^{k_{max}} k^2 q_k - \left(\sum_i^{k_{max}} k q_k \right)^2$$

There is a lot to unpack in this formula but basically r is the Pearson correlation coefficient between the degrees found at the two ends of the same link. As for Pearson correlation, $-1 \leq r \leq 1$ and if $r > 0$ the network is assortative, if $r < 0$ it is disassortative.

q_k is the probability that there is a degree k node at the end of the randomly selected link, sometimes called the distribution of the *remaining degree*. e_{ij} is the **degree correlation matrix**, the probability of finding a node with degrees i and j at the two ends of a randomly selected link. Note the potentially confusing indices here - usually we sum over nodes or edges, for the assortativity coefficient we sum over all possible values of degree.

References

- [1] Márton Pósfai and Albert-László Barabási. *Network science*. Cambridge University Press, 2016.
- [2] Mark Newman. *Networks*. Oxford university press, 2018.